

X-F SENSE: OPTIMAL SPATIO-TEMPORAL SENSITIVITY ENCODING FOR DYNAMIC MR IMAGING

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ABSTRACT

For high quality MR imaging of time-varying objects such as beating heart or brain hemodynamics, we need to reduce signal acquisition time without sacrificing the spatial resolution. Considerable efforts have been made to achieve this goal especially in the area of parallel imaging and temporal filtering techniques. Recently, researchers have tried to combine temporal filtering with parallel imaging so that any spatial residual artifact from parallel imaging can be further suppressed by temporal filtering. This paper extends the idea and proposes a new algorithm called the x-f SENSE (x-f domain SENSitivity Encoding) that optimally combines the parallel imaging and temporal filtering using the powerful lattice sampling theory. Unlike the conventional hybrid methods such as TSENSE and UNFOLD-SMASH, the x-f SENSE algorithm is optimal in the sense that it achieves the minimal sampling limit whereas the reconstruction procedure is as simple as that of the conventional SENSE. Theoretical analysis and simulation results demonstrate that x-f SENSE achieves theoretically optimal performance limit without any artifact.

1. INTRODUCTION

In the imaging of time-varying objects such as beating heart and brain hemodynamics, one of the main technical difficulties is overcoming the motion artifacts. Fast imaging sequence such as echo-planar imaging (EPI)[1] has been widely used in practise; however, EPI sacrifices the image quality to achieve high temporal resolution, hence new methods for improving both spatial and temporal resolution are needed.

To resolve this problem, considerable research efforts have been made in the area of parallel and dynamic imaging. Parallel imaging methods such as SMASH (SiMultaneous Acquisition of Spatial Harmonics)[2], SENSE (SENSitivity Encoding) [3], PILS (Partially Parallel Imaging with Localized Sensitivities)[4], and GRAPPA (GeneRalized Autocalibrating Partially Parallel Acquisitions)[5] reduce the scan time

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by skipping the phase encoding lines in the k-space, and the resultant aliasing artifacts are removed using spatial coil sensitivity encoding. Unlike the parallel imaging method, temporal filtering techniques such as UNFOLD (UNaliasing by Fourier-encoding the Overlaps Using the temporal Dimension)[6] exploit the temporal redundancies of time varying objects. More specifically, UNFOLD obtains the Fourier data on k-t-space in sheared grid pattern, in which the phase encoding in k-space is shifted at every frame. This results in the repetition of support region on x-f-space, and the original image can be reconstructed via x-f domain filtering.

Recently, researchers have tried to combine temporal filtering methods with parallel imaging techniques. For example, TSENSE [7] combines UNFOLD and SENSE so that any residual artifact from SENSE is temporally frequency-shifted to the band edge and thus may be further suppressed by temporal low-pass filtering. UNFOLD-SMASH [8] obtains the additional phase encoding lines using SMASH, after which the un-aliased images are obtained using temporal filtering. However, the method of optimizing the TSENSE has not been addressed, and the implementation of the UNFOLD-SMASH is prone to artifact since the error from interpolated k-space data can be propagated to the time and space.

In this paper, we propose a novel parallel imaging algorithm called the x-f SENSE (x-f domain SENSitivity Encoding). The x-f SENSE has many advantages over the conventional methods. First, x-f SENSE can be easily optimized using the lattice sampling theory so that the resultant MR signal acquisition can achieve the optimal spatial and temporal resolution tradeoff. Second, the acquisition procedure and reconstruction algorithm of x-f SENSE is so simple that it can be easily employed in practise. Therefore, we believe that our x-f SENSE can be used widely in practise.

2. LATTICE SAMPLING THEORY

A 2-dimensional lattice is defined as $\{n_1\mathbf{v}_1 + n_2\mathbf{v}_2 : n_1, n_2 \in \mathbb{Z}\}$ where $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent vectors. Here, $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2]$ denotes the sampling matrix or basis matrix, and the sampling density is defined by $1/d(\mathbf{V})$, where $d(\mathbf{V})$ denotes the determinant of a matrix $\mathbf{V}$. The reciprocal

lattice Λ^* is then defined as the lattice that has the basis matrix $\mathbf{V}^* = \mathbf{V}^{-1}$.

Let the signal $\{y(\mathbf{t}), \mathbf{t} \in \Lambda\}$ denote the discrete sampled signal of continuous signal $x(\mathbf{t})$ on lattice Λ . Using the 2-dimensional Fourier transform, the Fourier transform of $y(\mathbf{t})$ is given by

$$Y(\mathbf{f}) = \sum_{\mathbf{t} \in \Lambda} y(\mathbf{t}) (-j2\pi \mathbf{f}^T \mathbf{t}) d\mathbf{t}, \quad \mathbf{f} \in \mathbb{R}^2. \quad (1)$$

Now $Y(\mathbf{f})$ can be written as a function of $X(\mathbf{f})$, the Fourier transform of the continuous signal $x(\mathbf{t})$. More specifically, Fourier transform $Y(\mathbf{f})$ is composed of replicas on the reciprocal lattice Λ^* [9]:

$$Y(\mathbf{f}) = \frac{1}{d(\Lambda)} \sum_{\mathbf{r} \in \Lambda^*} X(\mathbf{f} + \mathbf{r}) \quad (2)$$

Let Λ' be another lattice. If every point of Λ' is a point of Λ , Λ' is a *sublattice* of Λ , and Λ is a *superlattice* of Λ' . If Λ' is a sublattice of Λ , the quotient $P = \frac{d(\Lambda)}{d(\Lambda')}$ is an integer and is called the *index* of Λ' in Λ . Then, we can easily show that the reciprocal lattice Λ'^* is a sublattice of Λ^* ; hence there exist P vectors $\mathbf{r}_i \in \Lambda^*$, $i = 1, \dots, P$ such that $\Lambda'^* = \cup_{i=1}^P (\Lambda^* + \mathbf{r}_i)$. Now define the *unit cell* of a lattice Λ' as a set $\mathcal{P} \subset \mathbb{R}^2$ such that $\mathbb{R}^2$ is the disjoint union of copies of $\mathcal{P}$ centered on each lattice point: $(\mathcal{P} + \mathbf{x}) \cap (\mathcal{P} + \mathbf{y}) = \emptyset$ for $\mathbf{x}, \mathbf{y} \in \Lambda'$, $\mathbf{x} \neq \mathbf{y}$, and $\cup_{\mathbf{x} \in \Lambda'} (\mathcal{P} + \mathbf{x}) = \mathbb{R}^2$. Then, Eq. (2) can be represented as

$$Y(\mathbf{f}) = \sum_{\mathbf{s} \in \Lambda'^*} Y(\mathbf{f} + \mathbf{s}), \quad (3)$$

where $Y(\mathbf{f})$ is given by

$$Y(\mathbf{f}) = \frac{1}{d(\Lambda')} \sum_{i=1}^P X(\mathbf{f} + \mathbf{r}_i), \quad \mathbf{f} \in \mathcal{P}. \quad (4)$$

3. DERIVATION OF X-F SENSE

MR signal measurement $s(k, t)$ in k -space at a given time t is related to the proton density $i(x, t)$ in a Fourier relationship

$$s(k, t) = \int_{-\infty}^{\infty} \alpha(x, t) i(x, t) e^{j2\pi kx} dx \quad (5)$$

where $\alpha(x, t)$ denotes the receiver coil sensitivity at the location x at a given time t . For parallel imaging, we usually assume that the coil sensitivity is time invariant [3], i.e. $\alpha(x, t) \simeq \alpha(x)$. Hence, $s(k, t)$ can be represented as a 2-D inverse Fourier transform

$$s(k, t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \alpha(x) I(x, f) e^{j2\pi(kx + tf)} dx df, \quad (6)$$

where $I(x, f)$ denotes Fourier transform of $i(x, t)$ along t . Now we use the vector notation $\mathbf{t} = (k, t)$ and $\mathbf{f} = (x, f)$ and the following definition

$$X(\mathbf{f}) = \alpha(x) I(x, f), \quad x(\mathbf{t}) = s(k, t). \quad (7)$$

Using Eqs. (4) and (7), the n -th coil image is given by

$$Y_n(\mathbf{f}) = \frac{1}{d(\Lambda)} \sum_{i=1}^P \alpha_n(\mathbf{f} + \mathbf{r}_i) I(\mathbf{f} + \mathbf{r}_i). \quad (8)$$

Now, we define the *coil sensitivity pattern* $\mathcal{K}$ and the *spectral repetition pattern*:

$$\mathcal{K} = \{\alpha_1, \dots, \alpha_K\}, \quad \mathcal{R} = \{\mathbf{r}_1, \dots, \mathbf{r}_P\}, \quad (9)$$

where K is the number of coils. Note that the index P corresponds to the number of overlap due to the k -t lattice sampling, hence the maximal achievable acceleration factor. Then, Eq. (8) can be written as a linear system

$$\mathbf{Y}(\mathbf{f}; \mathcal{K}) = \mathbf{W}(\mathcal{K}, \mathcal{R}) \mathbf{I}(\mathbf{f}; \mathcal{R}) \quad (10)$$

where

$$\mathbf{Y}(\mathbf{f}; \mathcal{K}) = [Y_1(\mathbf{f}), \dots, Y_K(\mathbf{f})]^T \quad (11)$$

$$\mathbf{I}(\mathbf{f}; \mathcal{R}) = [I(\mathbf{f} + \mathbf{r}_1), \dots, I(\mathbf{f} + \mathbf{r}_P)]^T \quad (12)$$

and the *aliasing structure matrix* $\mathbf{W}(\mathcal{K}, \mathcal{R})$ is given by

$$\mathbf{W}(\mathcal{K}, \mathcal{R}) = \frac{1}{P} \begin{bmatrix} \alpha_1(\mathbf{f} + \mathbf{r}_1) & \alpha_1(\mathbf{f} + \mathbf{r}_2) & \dots & \alpha_1(\mathbf{f} + \mathbf{r}_P) \\ \alpha_2(\mathbf{f} + \mathbf{r}_1) & \alpha_2(\mathbf{f} + \mathbf{r}_2) & \dots & \alpha_2(\mathbf{f} + \mathbf{r}_P) \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_K(\mathbf{f} + \mathbf{r}_1) & \alpha_K(\mathbf{f} + \mathbf{r}_2) & \dots & \alpha_K(\mathbf{f} + \mathbf{r}_P) \end{bmatrix} \quad (13)$$

Note that for general x - f spectrum support, Eq. (10) has the unique solution for $K \geq P$, hence the maximum K times accelerated acquisition of MR signal is feasible using K phase array coils. For dynamically varying objects, the spatio-temporal support is usually band-limited as shown in Figs. 4(a)(b), and further acceleration of MR signal acquisition is possible. To show this, let us define the following set

$$\mathcal{M} = \{\mathbf{r}_i \in \Lambda^* : I(\mathbf{f} + \mathbf{r}_i) \text{ is non-zero}\}. \quad (14)$$

For the band-limited x - f support the number of nonzero elements of the vector $\mathbf{I}(\mathcal{R}; \mathbf{f})$ is usually less than P , i.e. $|\mathcal{M}| \leq P$, where $|\cdot|$ denotes the cardinality of a set. Hence, Eq. (10) can be represented as

$$\mathbf{Y}(\mathbf{f}; \mathcal{K}) = \mathbf{W}(\mathcal{K}, \mathcal{M}) \mathbf{I}(\mathbf{f}; \mathcal{M}). \quad (15)$$

Note that the set $\mathcal{M}$ in (14) is dependent on specific x - f space position $\mathbf{f}$. Therefore, in order to guarantee the perfect reconstruction of x - f SENSE, Eq. (15) should have the unique solution for every $\mathcal{M}$. This can be characterized by the universality of the coil sensitivity pattern [10]. A coil sensitivity pattern $\mathcal{K}$ is called universal if the matrix $\mathbf{W}(\mathcal{K}, \mathcal{M})$ has full column rank for every set $\mathcal{M}$ of $|\mathcal{M}| < |\mathcal{K}|$. Assuming the coil sensitivity pattern is universal, we can obtain the following perfect reconstruction condition for x - f SENSE:

Theorem 3.1. Let $K = |\mathcal{K}|$ and $M = |\mathcal{M}|$. If coil sensitivity pattern $\mathcal{K}$ is universal and $M \leq K$ for all $\mathbf{f} \in \mathcal{P}$, the perfect image reconstruction is possible using x -f SENSE.

One of the important problem in parallel and dynamic MR imaging is understanding the fundamental limit of the performance enhancement. This problem can be addressed within the framework of the non-uniform sampling theory [10]. Here, the minimal sampling rate for perfect signal reconstruction is given by the so-called Landau limit:

$$r_{min} = \frac{Area(\mathcal{S})}{Area(FOV)} \quad (16)$$

where $Area(\mathcal{S})$ and $Area(FOV)$ denote the Lesbeque measures of spectral support in k - t space and FOV, respectively. We can show that the maximal acceleration factor using x -f SENSE can be related to the Landau as following:

Theorem 3.2. The acceleration factor we can achieve using x -f SENSE is given by

$$P \leq \frac{K}{r_{min}} \quad (17)$$

where K denotes the number of RF coils. Furthermore, the upper bound is an achievable bound if $|\mathcal{K}| = |\mathcal{M}|$ for all $\mathbf{f} \in \mathcal{U}$.

4. NUMERICAL EXAMPLES

A numerical simulation is conducted to demonstrate the effectiveness of x -f SENSE. The simulation phantom is made of 3 static and 2 time varying ellipses similar to [11]. The time-varying component of the phantom consists of two centered ellipse, a and b , which are at the right angle to each other, and whose proton densities are temporally modulated by the sinc function $\text{sinc}(\pi t)$ and $\text{sinc}[\pi(t - 2)]$, respectively, shown in Fig. 1. The size of images at each time instant is 128×128 pixels and we sample 128 frames along time axis. Among 128 time frames, 9 frames are selected and displayed in Fig. 2. We use two coils for the simulation of parallel

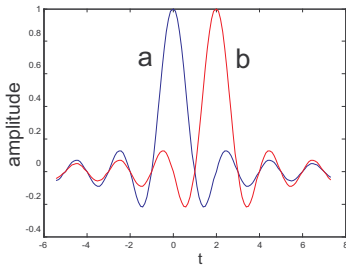


Fig. 1. Temporal modulation of a and b densities.

data acquisition. The sensitivity maps of coil 1 and coil 2 are given by Figs. 3(a)(b), respectively. In order to obtain the

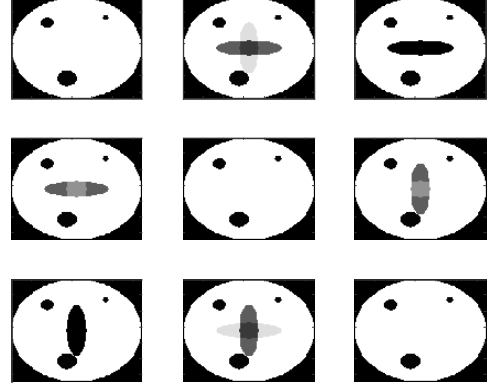


Fig. 2. The 45th, 50th, 55th, 60th, 65th, 70th, 75th, 80th, and 85th time frames of the original synthetic phantom.

optimal sampling pattern in k - t space, the spectral support of the phantom is computed using the Fourier transform of the phantom along x and t axis. The resultant spectral supports of the phantom in x - f space are given in Fig. 4(a)(b).¹ Note that in Fig. 4(a)(b), the actual x - f spectral support of the phantom is dependent on the y -values. Hence, in order to prevent aliasing in any y , the maximum bounding box for the x - f spectral support for all y -value is obtained and used for sampling pattern design. Since we use two coils ($K = 2$), the number of non-zero overlap due to the aliasing should not be bigger than twice. Based on the simple calculation, we found that the optimal sampling lattice in k - t domain can be given by the following basis matrix $\mathbf{V}$.

$$\mathbf{V} = \begin{pmatrix} 2 & 0 \\ 8 & 16 \end{pmatrix} \quad (18)$$

where the numbers are normalized in pixel unit. Note that the acceleration factor $P = 32$ can be achieved using the basis matrix Eq. (18). While the number of overlap due to the aliasing is $P = 32$, the actual number of *nonzero* overlap is $M = 2$. Therefore, the perfect reconstruction is possible using two coils. For the x - f spectral support given in Figs. 4(a)(b), the corresponding spectrum replication patterns from the sampling by Eq. (18) are given in Figs. 4(c)(d). Note that the number of overlap for each x - f sampling point is dependent on y coordinate. Since we choose the maximal bounding box for the x - f support to optimize the sampling pattern, we can guarantee that the number of overlap for each x - f sampling for all y -coordinate should be either 1 or 2. Hence, by solving Eq. (15) for $K = 2, M = 1$ or $K = 2, M = 2$, the perfect reconstruction is possible. The reconstruction results are illustrated in Fig. 5.

¹Since the MR signal samples along the x axis are obtained simultaneously during one echo period, we only need to consider x - f spectral supports.

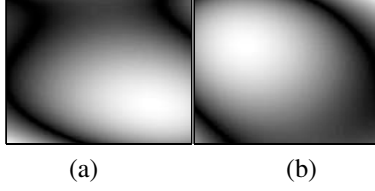


Fig. 3. The sensitivity maps of (a) coil 1 and (b) coil 2.

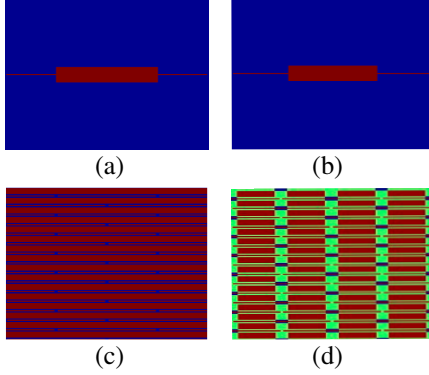


Fig. 4. The spectral supports at (a) $y = 64$ and (b) $y = 60$, and the corresponding aliasing patterns (c) and (d), respectively.

5. CONCLUSION

This paper proposes a novel x-f SENSE algorithm that optimally combines the parallel imaging and temporal filtering using the powerful lattice sampling theory. Unlike the conventional hybrid methods such as TSENSE and UNFOLD-SMASH, x-f SENSE is optimal in the sense it achieves the maximum acceleration factor. Furthermore, the reconstruction procedure of x-f SENSE is simple and can be easily implemented. Theoretical results and numerical simulation demonstrated that x-f SENSE is very powerful parallel imaging method for time varying objects. Therefore, we believe that x-f SENSE may be very useful in many clinical applications.

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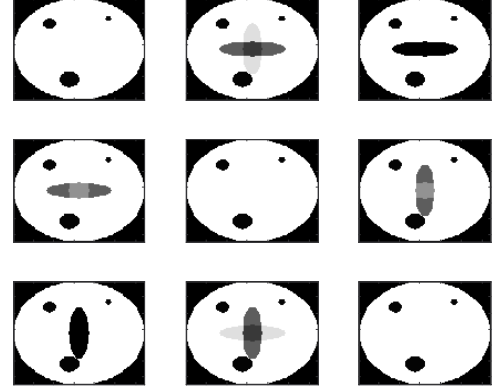


Fig. 5. Reconstruction results using x-f SENSE.

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